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MINI PROJECT

LIBRARY INFORMATION SYSTEM TYPES

OF COMMANDS:

DDL[DATA DEFINITION LANGUAGE]

1.CREATE:

BOOKSS:

```
SQL> select*from BOOKSS;
```

AUTHOR	ISBN	TITLE	EDITION	CATEGORY	PRICE
silber	10101	database	7	CSE	25000
samantha	20202	ds	5	CSE	35000
albert	24170	os	6	IT	10000

READERS:

```
SQL> select*from READERS;
```

USER_ID	EMAIL	ADDRESS	PHONE_NO	NAME
1010	t@gmail	pondy	1317	thanu
2003	r@gmail	chennai	2431	ramya
2002	d@gmail	kerala	3113	dina

PUBLISHERSS:

```
SQL> select*from PUBLISHERSS;
```

PUBLISHER_	YEAR_OF_PUBLICATION	NAME
1724	2004	karthi
1106	1995	abhi
6060	2000	janu

AUTHENTICATION:

```
SQL> select*from AUTHENTICATION;
```

LOGIN_ID	PASSWORD
hema	hema24
sana	sana17
ashwin	ashwin04

REPORTS:

```
SQL> select*from REPORTS;
```

USER_ID	REG_NO	BOOK_NO	ISSUE_DATE
2024	21td0065	1317	24/10/2023
2526	22td0010	2431	10/01/2023
8691	19td0015	3544	06/11/2023

STAFF:

```
SQL> select*from STAFF;
```

NAME	STAFF_ID
hemalatha	hema
pugal	riya
mohan	jaivanth

2.ALTER:

```
SQL> alter table TEACHERS add(area char(10));
```

Table altered.

```
SQL> select*from TEACHERS;
```

NAME	STAFF_ID	AREA
hemalatha	hema	
pugal	riya	
mohan	jaivanth	
varshini	vasu	

3.DROP:

```
SQL> drop table REPORTS;
```

Table dropped.

```
SQL> select*from REPORTS;
```

```
select*from REPORTS
*
```

ERROR at line 1:
ORA-00942: table or view does not exist

4.TRUNCATE:

```
SQL> truncate table PUBLISHERSS;  
  
Table truncated.  
  
SQL> select*from PUBLISHERSS;  
  
no rows selected
```

5.RENAME:

```
SQL> rename STAFF to TEACHERS;  
  
Table renamed.  
  
SQL> select*from TEACHERS;
```

NAME	STAFF_ID
hemalatha	hema
pugal	riya
mohan	jaivanth
varshini	vasu

DML[DATA MANIPULATION LANGUAGE]

1.SELECT:

```
SQL> select*from BOOKSS;
```

AUTHOR	ISBN	TITLE	EDITION	CATEGORY	PRICE
silber	10101	database	7	CSE	25000
samantha	20202	ds	5	CSE	35000
albert	24170	os	6	IT	10000

2.INSERT:

```
SQL> Insert into STAFF(name,staff_id)values('varshini','vasu');
```

```
1 row created.
```

```
SQL> select*from STAFF;
```

NAME	STAFF_ID
hemalatha	hema
pugal	riya
mohan	jaivanth
varshini	vasu

3.UPDATE:

```
SQL> update BOOKSS set price='15000' where author='albert';
```

```
1 row updated.
```

```
SQL> select*from BOOKSS;
```

AUTHOR	ISBN	TITLE	EDITION	CATEGORY	PRICE
silber	10101	database	7	CSE	25000
samantha	20202	ds	5	CSE	35000
albert	24170	os	6	IT	15000

4.DELETE:

```
SQL> delete from PUBLISHERSS where name='karthi';

1 row deleted.

SQL> select*from PUBLISHERSS;

PUBLISHER_ YEAR_OF_PUBLICATION  NAME
-----
1106        1995                abhi
6060        2000                janu
```

TRANSACTION CONTROL COMMMANDS:

1.DESCRPTION:

```
SQL> desc REPORTS;
Name                                     Null?   Type
-----
USER_ID                                VARCHA2(10)
REG_NO                                VARCHA2(20)
BOOK_NO                                VARCHA2(10)
ISSUE_DATE                             VARCHA2(15)
```

2.COMMIT:

```
SQL> commit;

Commit complete.

SQL> select*from PUBLISHERSS;

PUBLISHER_ YEAR_OF_PUBLICATION  NAME
-----
1106        1995                abhi
6060        2000                janu
```

3.SAVEPOINT:

```
SQL> savepoint sana;

Savepoint created.

SQL> select*from AUTHENTICATION;
```

LOGIN_ID	PASSWORD
hema	hema24
sana	sana17
ashwin	ashwin04

4.ROLLBACK:

```
SQL> delete from BOOKSS where author='samantha';

1 row deleted.

SQL> select*from BOOKSS;
```

AUTHOR	ISBN	TITLE	EDITION	CATEGORY	PRICE
silber	10101	database	7	CSE	25000
albert	24170	os	6	IT	15000

```
SQL> roll back;
Rollback complete.
SQL> select*from BOOKSS;
```

AUTHOR	ISBN	TITLE	EDITION	CATEGORY	PRICE
silber	10101	database	7	CSE	25000
samantha	20202	ds	5	CSE	35000
albert	24170	os	6	IT	15000

ARITHMETIC OPERATORS

1.MAX:

```
SQL> select max(price)from BOOKSS;  
  
MAX(PRICE)  
-----  
35000
```

2.MIN:

```
SQL> select min(price)from BOOKSS;  
  
MIN(PRICE)  
-----  
15000
```

3.COUNT:

```
SQL> select count(name)from STAFF;  
  
COUNT(NAME)  
-----  
4
```


4.AVERAGE:

```
SQL> select avg(price)from BOOKSS;

AVG(PRICE)
-----
      25000
```

5.SUM:

```
SQL> select sum(price)from BOOKSS;

SUM(PRICE)
-----
      75000
```

SET OPERATORS:

UNION:

```
SQL> select user_id from READERS union select user_id from REPORTS;

USER_ID
-----
1010
2002
2003
2024
2526
8691

6 rows selected.
```

UNION ALL:

```
SQL> select login_id from AUTHENTICATION union all select staff_id from STAFF;

LOGIN_ID
-----
hema
sana
ashwin
hema
riya
jaivanth
vasu

7 rows selected.
```

MINUS:

```
SQL> select login_id from AUTHENTICATION minus select staff_id from STAFF;

LOGIN_ID
-----
ashwin
sana
```

INTERSECT:

```
SQL> select login_id from AUTHENTICATION intersect select staff_id from STAFF;

LOGIN_ID
-----
hema
```

DATA QUERY LANGUAGE:

1.SELECT:

```
SQL> select distinct title from BOOKSS;

TITLE
-----
os
database
ds
```

2.ORDER BY CLAUSE:

```
SQL> select email from READERS where name='dina' order by email;

EMAIL
-----
d@gmail

SQL> select * from PUBLISHERSS order by year_of_publication desc,publisher_id asc;

PUBLISHER_ YEAR_OF_PUBLICATION  NAME
-----
6060      2000      janu
1106      1995      abhi
```

3.GROUP BY CLAUSE:

```
SQL> select name from STAFF group by name;

NAME
-----
varshini
pugal
mohan
hemalatha

SQL> select author,avg(price) as averageprice from BOOKSS group by author;

AUTHOR      AVERAGEPRICE
-----
albert      15000
silber      25000
samantha    35000
```